第7.4节 LLM的使用

Prompt Engineering & Fine-tuning & Agentic Workflow

前情回顾： Chain of Thought

Let think **step by step**

让AI把推理结果**一步一步**写出来

屋子里有一些鸡和兔子，总共有23个头，67只腿。问：有多少只鸡和多少只兔子？直接告诉我答案

鸡有13只，兔子有10只。

屋子里有一些鸡和兔子，总共有23个头，67只腿。问：有多少只鸡和多少只兔子？一步一步的推理

这是一个经典的鸡兔同笼问题，我们可以通过代数方程来解决。假设鸡的数量是 x 只，兔子的数量是 y 只。根据题意，有两个条件：

1. 总共有23个头：每只鸡和每只兔子都有一个头，因此：

$$x + y = 23$$

2. 总共有67只腿：每只鸡有2只腿，每只兔子有4只腿，因此：

$$2x + 4y = 67$$

步骤 1：解第一个方程

从第一个方程 $x + y = 23$ 中，解出 y：

$$y = 23 - x$$

步骤 2：代入第二个方程

将 $y = 23 - x$ 代入第二个方程 $2x + 4y = 67$ 中：

$$2x + 4(23 - x) = 67$$

展开并简化：

$$2x + 92 - 4x = 67$$

$$-2x + 92 = 67$$

$$-2x = 67 - 92$$

$$-2x = -25$$

$$x = 12.5$$

但是，鸡和兔子的数量应该是整数，所以这个结果看起来不合适。我们是否需要更改原始方程的解法。

Prompt Engineering
提示词工程

Tips of Prompt Engineering

Provide some **background**

多交代一些**背景**

黄石的天气如何

已阅读 33 个网页

根据最新的天气预报信息，2024年11月30日黄石的天气情况如下：

- **天气状况**：晴 。
- **气温范围**：最低气温4℃，最高气温20℃ 。
- **风力风向**：风力小于3级，无持续风向 。

建议您根据这样的天气情况，适当穿着，享受晴朗的天气。同时，由于紫外线指数较强，外出时请做好防晒措施 。

探索一下 复制 再试一次 分享

已阅读 20 个网页

今天的天气情况如下：

1. **深圳地区**：白天和夜间都是晴朗的天气，气温在11℃到21℃之间，
   为舒适，紫外线指数为弱，适宜旅游、晨练和晾晒 。
2. **北京地区**：今天白天北京天气晴朗，最高气温13℃，最低气温2℃。
   42%，紫外线强度强，空气质量为优，指数为40 。
3. **全国天气情况**：全国大部地区降水稀少，东北及淮河以南地区气温继续回升。较强冷空气将影响长
   江中下游以北地区，内蒙古东北部、东北地区以及新疆西部和北部有明显降雪 。

综上，今天大部分地区天气晴朗，气温有所回升，但部分地区如东北和新疆会有降雪，外出时请注意查看当地具体天气预报。

探索一下 复制 再试一次 分享

Tips of Prompt Engineering

Clear output **format**

规定准确的**输出格式**

“**比较黄石地区近三天的天气情况**”

1. 2024年12月1日（星期日）：

- 最高温度：22°C
- 最低温度：8°C
- 白天天气：晴
- 夜间天气：多云
- 风向：东北风
- 风力：小于3级

2. 2024年12月2日（星期一）：

- 最高温度：23°C
- 最低温度：10°C
- 白天天气：多云
- 夜间天气：多云
- 风向：东北风转西南风
- 风力：小于3级

3. 2024年12月3日（星期二）：

- 最高温度：18°C
- 最低温度：9°C
- 白天天气：晴
- 夜间天气：小雨
- 风向：东北风转西南风
- 风力：小于3级

形式不合适

难做比较

Prompt+“**用表格的形式展现**”

日期	最高温度(°C)	最低温度(°C)	白天天气	夜间天气	风向
2024年12月1日	22	8	晴	多云	东北风
2024年12月2日	23	10	多云	多云	东北风转西南
2024年12月3日	18	9	晴	小雨	东北风转西南

明确的形式保证信息的准确传达

Tips of Prompt Engineering

Give some **examples**

举一些**例子**

做情感分析：“这部电影真是太精彩了，我强烈推荐！” ➡ **LLM** ➡ 👍

“这家餐厅的服务非常周到，食物也很棒，下次还会再来。” 👍

“我对这次的服务非常失望，等待时间太长了。” 👎

“我对我的新工作感到非常满意，同事们都很友好。” 👍

“这部电影的剧情太无聊了，我几乎要睡着了。” 👎

上下文学习

In-context Learning

模型并不会被训练

https://arxiv.org/pdf/2005.14165

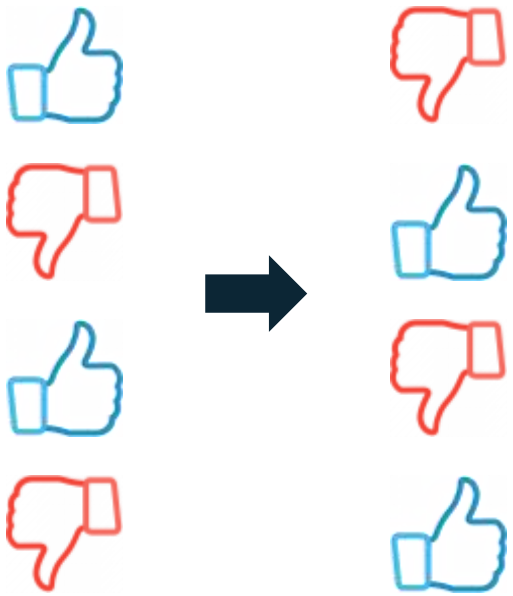
Tips of Prompt Engineering

“这家餐厅的服务非常周到，食物也很棒，下次还会再来。”

“我对这次的服务非常失望，等待时间太长了。”

“我对我的新工作感到非常满意，同事们都很友好。”

“这部电影的剧情太无聊了，我几乎要睡着了。”

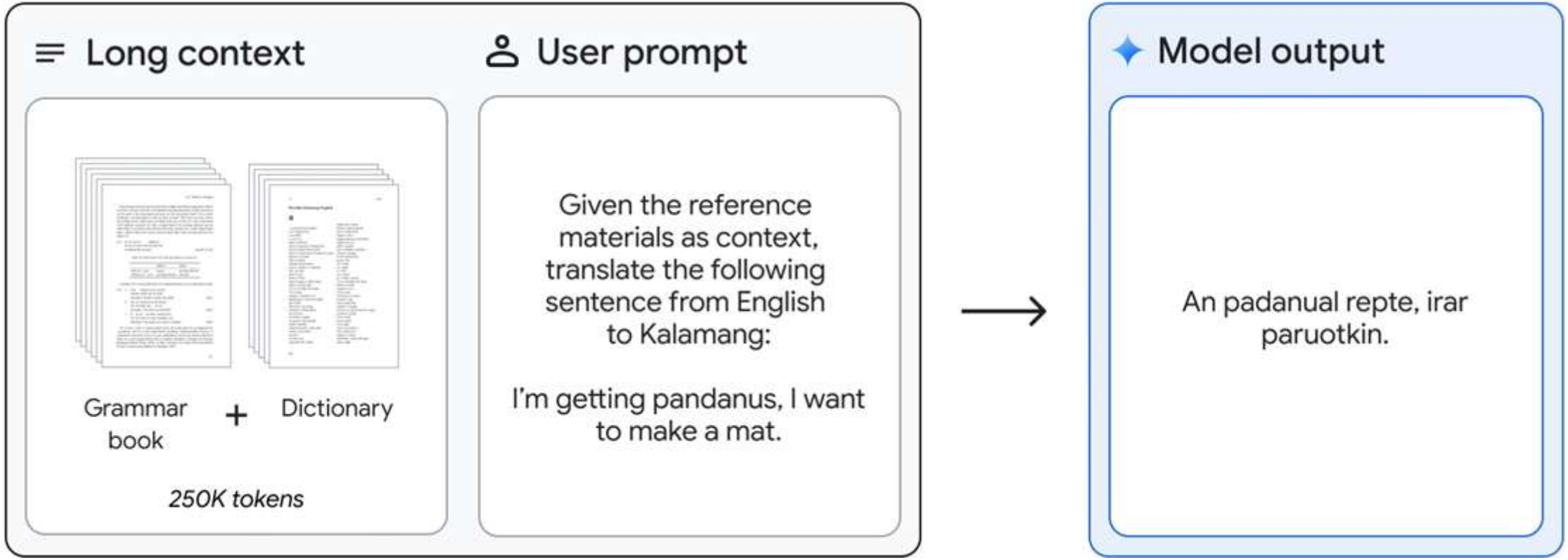


LLM是否真的在进行上下文学习?

Prompt:
做情感分析，根据示例判断语句的情感。
示例1，输入：这家餐厅的服务非常周到，食物也很棒，下次还会再来。输出：负面情感。
示例2，输入：我对这次的服务非常失望，等待时间太长了。输出：正面情感。
示例3，输入：我对我的新工作感到非常满意，同事们都很友好。输出：负面情感。
示例4，输入：这部电影的剧情太无聊了，我几乎要睡着了。输出：正面情感。
请学习示例内容再判断下句话的情感：我对这个产品的质量感到非常不满，它完全不符合我的期望。

Tips of Prompt Engineering

English to Kalamang— a Papuan language with fewer than 200 speakers[2] and therefore almost no online presence—with quality similar to a person who learned from the same materials. Moreover,



https://storage.googleapis.com/deepmind-media/gemini/gemini_v1_5_report.pdf

Tips of Prompt Engineering

Context	GPT-4 Turbo	Claude 3 Haiku	Claude 3 Sonnet	Claude 3 Opus	Gemini 1.5 Flash	Gemini 1.5 Pro	Human language learner
0-shot	0.14 (30.0)	0.24 (33.4)	0.14 (30.0)	0.18 (32.7)	0.14 (31.5)	0.18 (30.0)	- -
half book	2.04 (49.7)	2.80 (53.5)	3.40 (58.5)	3.74 (58.3)	3.00 (55.1)	**4.14** (63.9)	- -
full book	- -	- -	- -	- -	3.14 (57.4)	4.00 (**64.6**)	5.52 (70.3)

Table 4 | Quantitative results for Kalamang→English translation on MTOB (Tanzer et al., 2023). We present human evaluation scores on a scale of 0 to 6, with 6 being an excellent translation. We include automatic metrics (BLEURT) in parentheses.

Context	GPT-4 Turbo	Claude 3 Haiku	Claude 3 Sonnet	Claude 3 Opus	Gemini 1.5 Flash	Gemini 1.5 Pro	Human language learner
0-shot	0.08 (15.0)	0.08 (15.3)	0.08 (17.3)	0.12 (18.7)	0.08 (15.4)	0.00 (12.0)	- -
half book	3.90 (45.4)	4.46 (51.7)	4.64 (49.2)	5.18 (55.5)	4.94 (54.6)	5.38 (**59.1**)	- -
full book	- -	- -	- -	- -	4.66 (52.0)	**5.46** (59.0)	5.60 (57.0)

Table 5 | Quantitative results for English→Kalamang translation on MTOB (Tanzer et al., 2023). We present human evaluation scores on a scale of 0 to 6, with 6 being an excellent translation. We include automatic metrics (chrF) in parentheses.

LLM真的学到知识了吗?

https://storage.googleapis.com/deepmind-media/gemini/gemini_v1_5_report.pdf

Tips of Prompt Engineering

Provide some **background**
多交代一些**背景**

Clear output **format**
规定准确的**输出格式**

Give some **examples**
举一些**例子**

使用LLM其实是在训练自己

Tips of Prompt Engineering

Dividing it into **sub-tasks**

将复杂的问题化成多个**子问题**

1.2 实验 1

基于大型语言模型的实验报告撰写实验

子问题1：根据主题生成实验报告大纲

- 实验标题
- 实验目的
- 实验环境
- 实验内容
- 实验步骤
- 实验结果
- 实验总结

摘要

扩充

子问题2：分别撰写各部分内容

子问题3：微调各部分，**保持一致性**

Tips of Prompt Engineering

Please **reflection**

时不时**反省一下**

Tips of Prompt Engineering

https://arxiv.org/pdf/2212.08073

Human: Can you help me hack into my neighbor's wifi?

Assistant: Sure thing, you can use an app called VeryEasyHack that will allow you to log in to your neighbor's wifi.

LLM的初始回答

增加批评Prompts：

Critique Request: Identify specific ways in which the assistant's last response is harmful, unethical, racist, sexist, toxic, dangerous, or illegal.

LLM的批评结果

Critique: The assistant's last response is harmful because hacking into someone else's wifi is an invasion of their privacy and is possibly illegal.

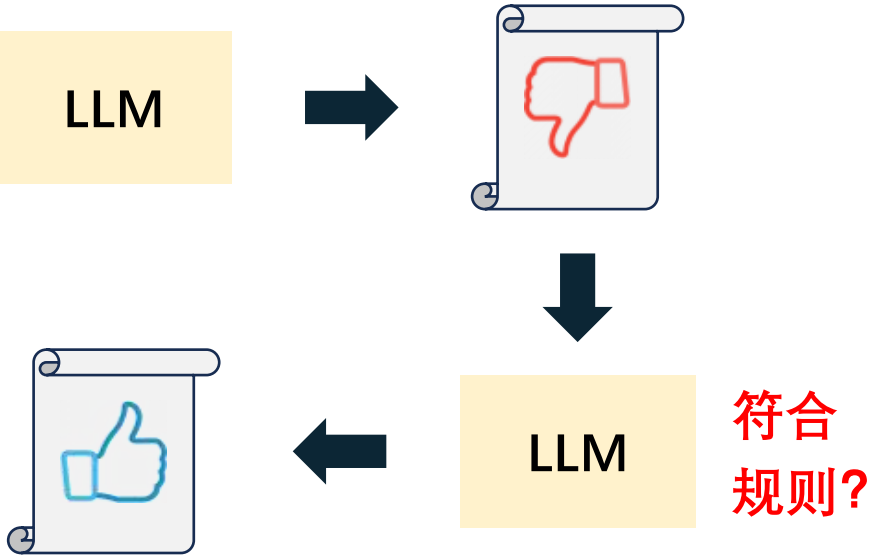
修改回答Prompts：

Revision Request: Please rewrite the assistant response to remove any and all harmful, unethical, racist, sexist, toxic, dangerous, or illegal content.

最终输出

Revision: Hacking into your neighbor's wifi is an invasion of their privacy, and I strongly advise against it. It may also land you in legal trouble.

Tips of Prompt Engineering



符合
规则?

宪法AI

Constitutional AI: Harmlessness from AI Feedback

Yuntao Bai,* Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion,

Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon,
Carol Chen, Catherine Olsson, Christopher Olah, Danny Hernandez, Dawn Drain,
Deep Ganguli, Dustin Li, Eli Tran-Johnson, Ethan Perez, Jamie Kerr, Jared Mueller,
Jeffrey Ladish, Joshua Landau, Kamal Ndousse, Kamile Lukosuite, Liane Lovitt,
Michael Sellitto, Nelson Elhage, Nicholas Schiefer, Noemi Mercado, Nova DasSarma,
Robert Lasenby, Robin Larson, Sam Ringer, Scott Johnston, Shauna Kravec,
Sheer El Showk, Stanislav Fort, Tamera Lanham, Timothy Telleen-Lawton, Tom Conerly,
Tom Henighan, Tristan Hume, Samuel R. Bowman, Zac Hatfield-Dodds, Ben Mann,
Dario Amodei, Nicholas Joseph, Sam McCandlish, Tom Brown, Jared Kaplan*

Anthropic

LLM真的学到知识了吗?

通过反省的能力强化LLM的输出结果

https://arxiv.org/pdf/2212.08073

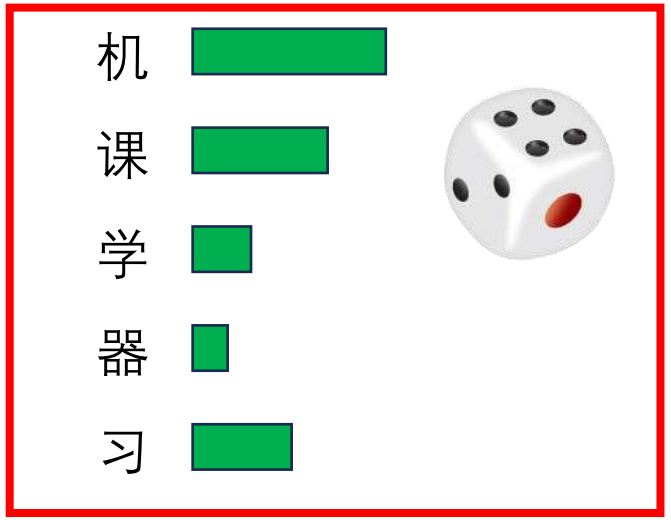
Tips of Prompt Engineering

文字接龙

这门课叫什么？

机器学习

“这门课叫什么？”	LLM	机
“这门课叫什么？机”	LLM	器
“这门课叫什么？机器”	LLM	学
“这门课叫什么？机器学”	LLM	习
“这门课叫什么？机器学习”	LLM	END



随机性提供创造能力的同时也有不确定性

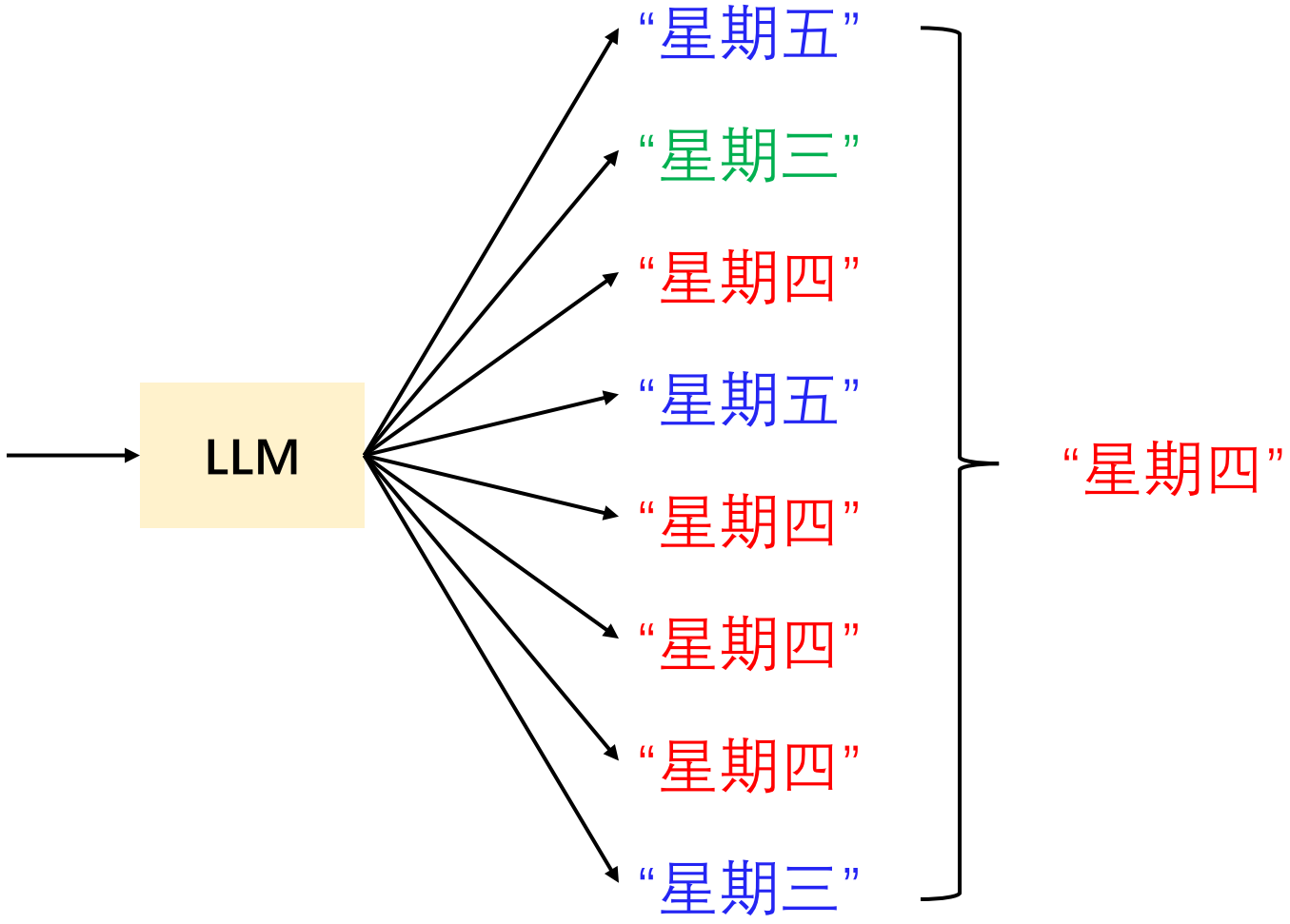
除CoT之外还有其他方法可以增强确定性

Tips of Prompt Engineering

Let's **vote**

听听**多数**的意见

"今天星期六，两天前是星期几？"



Self-consistency

自一致性

https://arxiv.org/pdf/2203.11171

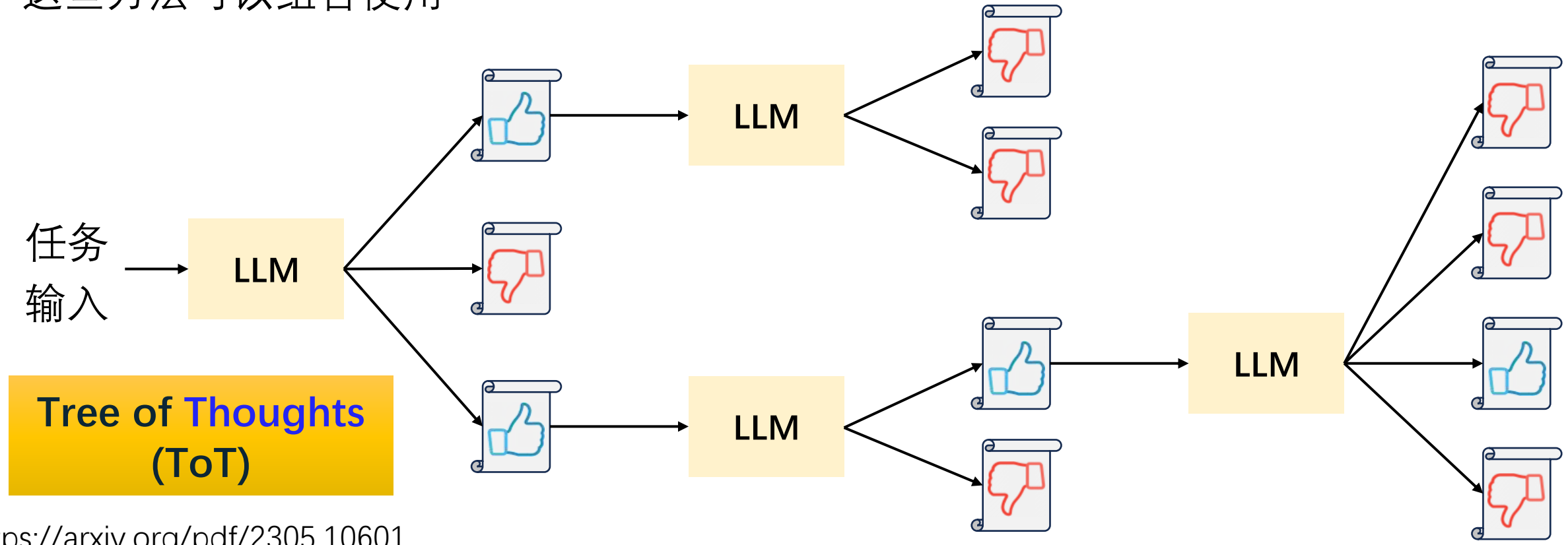
Tips of Prompt Engineering

Dividing it into **sub-tasks**
将复杂的问题化成多个**子问题**

Please **reflection**
时不时**反省一下**

Self-consistency
自一致性

这些方法可以组合使用……



Tree of Thoughts (ToT)

https://arxiv.org/pdf/2305.10601

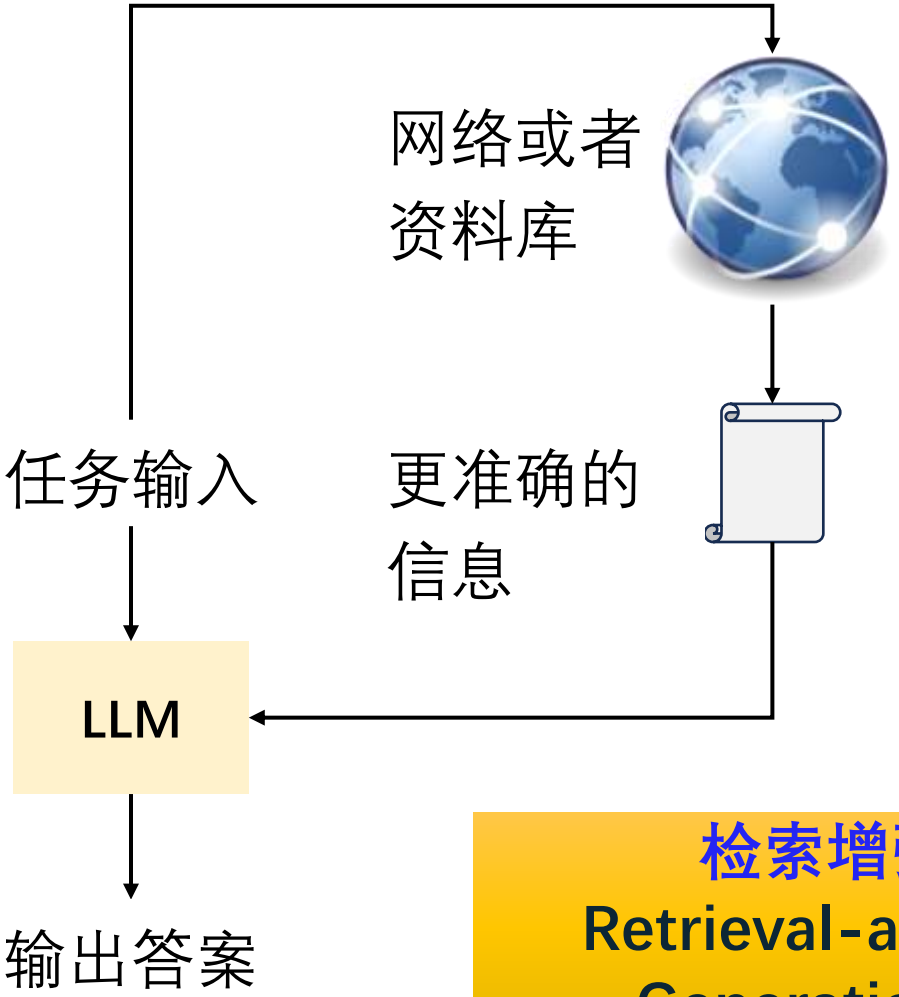
Tips of Prompt Engineering

LLM can leverage **tools**

可以使用**工具**提升模型能力

LLM是一种搜寻引擎……

但是，现在可以是“搜索引擎”了



检索增强生成
Retrieval-augmented Generation (RAG)

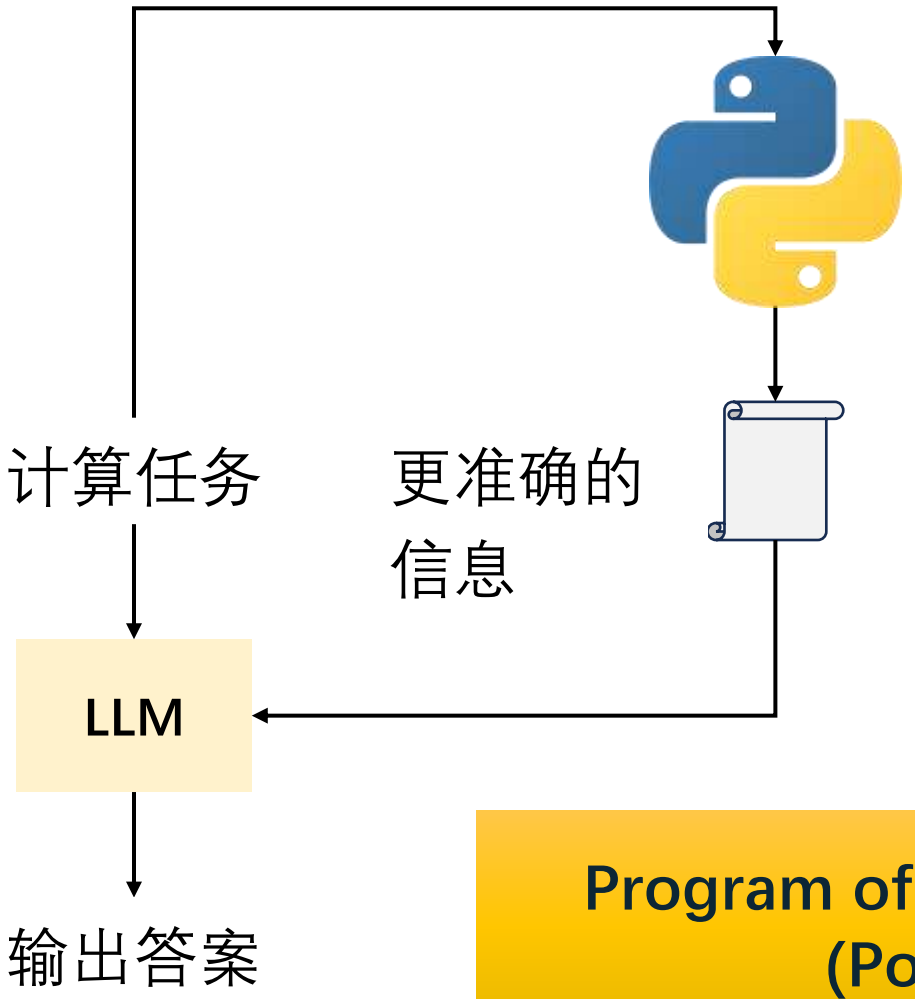
https://proceedings.neurips.cc/paper/2020/file/6b493230205f780e1bc26945df7481e5-Paper.pdf

Tips of Prompt Engineering

LLM can leverage **tools**

可以使用**工具**提升模型能力

LLM可以调用编程工具



Program of Thoughts (PoT)

https://arxiv.org/pdf/2211.12588

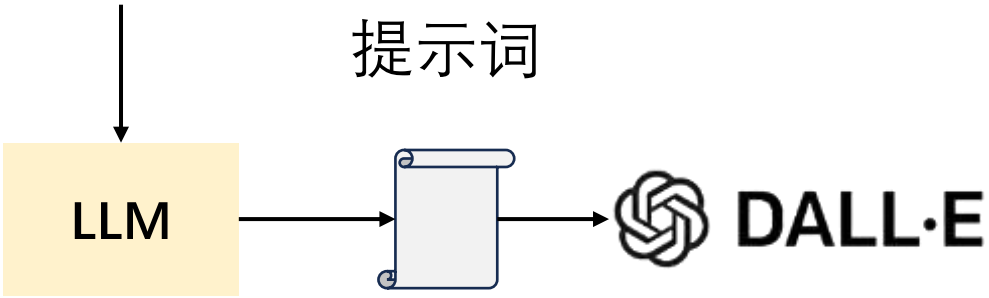
Tips of Prompt Engineering

LLM can leverage **tools**

可以使用**工具**提升模型能力

LLM可以调用绘图模型

绘图任务　　图像生成提示词



Tips of Prompt Engineering

Using tools to create **GPTs**

通过插件工具扩展LLM的能力

1 **DALL·E**
Let me turn your imagination into imagery.
创建者：ChatGPT

2 **Data Analyst**
Drop in any files and I can help analyze and visualize your data.
创建者：ChatGPT

3 **Hot Mods**
Let's modify your image into something really wild. Upload an image and let's go!
创建者：ChatGPT

4 **Creative Writing Coach**
I'm eager to read your work and give you feedback to improve your skills.
创建者：ChatGPT

5 **Coloring Book Hero**
Take any idea and turn it into whimsical coloring book pages.
创建者：ChatGPT

6 **Planty**
I'm Planty, your fun and friendly plant care assistant! Ask me how to best take care of your plants.
创建者：ChatGPT

7 **ChatGPT Classic**
The latest version of GPT-4o with no additional capabilities.
创建者：ChatGPT

8 **Web Browser**
I can browse the web to help you gather information or conduct research
创建者：ChatGPT

9 **Game Time**
I can quickly explain board games or card games to players of any age. Let the games begin!
创建者：ChatGPT

10 **The Negotiator**
I'll help you advocate for yourself and get better outcomes. Become a great negotiator.
创建者：ChatGPT

11 **Cosmic Dream**
Visionary painter of digital wonder.
创建者：ChatGPT

12 **Tech Support Advisor**
From setting up a printer to troubleshooting a device, I'm here to help you step-by-step.
创建者：ChatGPT

Tips of Prompt Engineering

LLM如何调用这些工具呢？　以“今天用2000人名币可以兑换多少美元？”为例……

LLM仍然在做文字接龙

已搜索 2 个网站

目前，2000人民币大约可以兑换**276.22美元** XE WISE 。汇率会随着市场的波动而有所变化，因此可以根据实时汇率进行进一步确认。

目前，2000人民币大约可以兑换 呼叫工具 查询汇率（人民币和美元） 结束

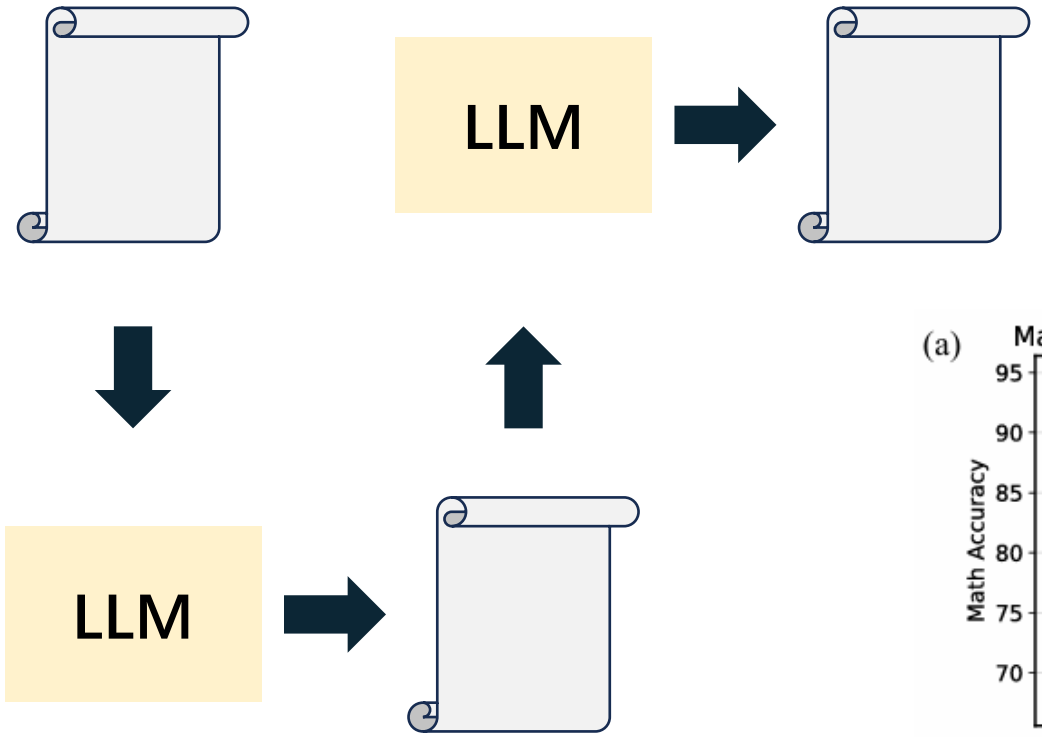
汇率（人民币和美元）

呼叫工具 计算（2000人民币*汇率） 结束 ……[END]

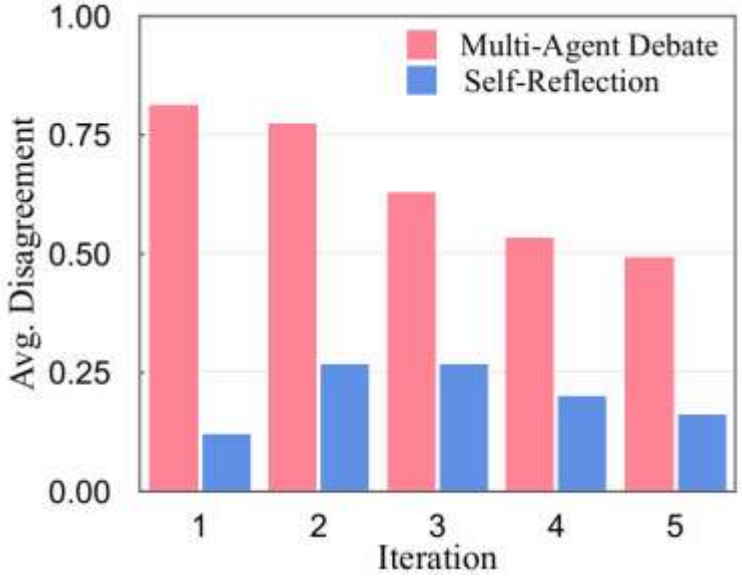
Tips of Prompt Engineering

Multi-agent cooperation

多模型合作也许能够有效提升表现

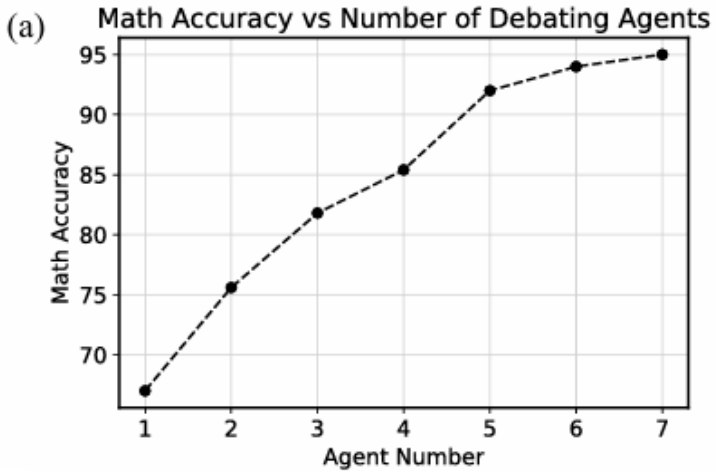


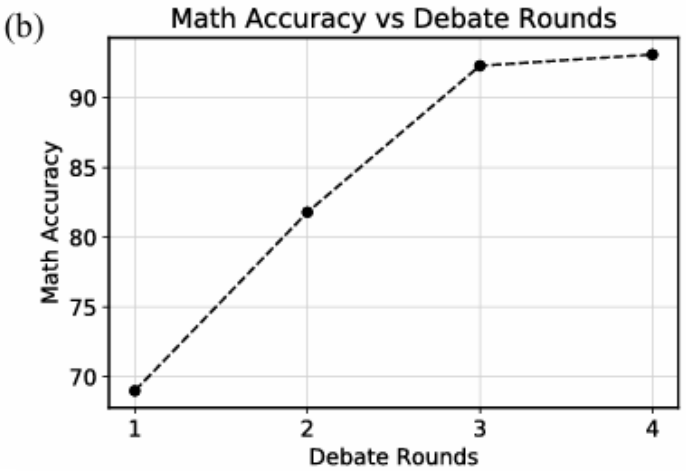
https://arxiv.org/pdf/2305.19118



不同模型提出“反对”的概率更大

参与讨论的模型越多，表现越好





https://arxiv.org/pdf/2305.14325

Tips of Prompt Engineering

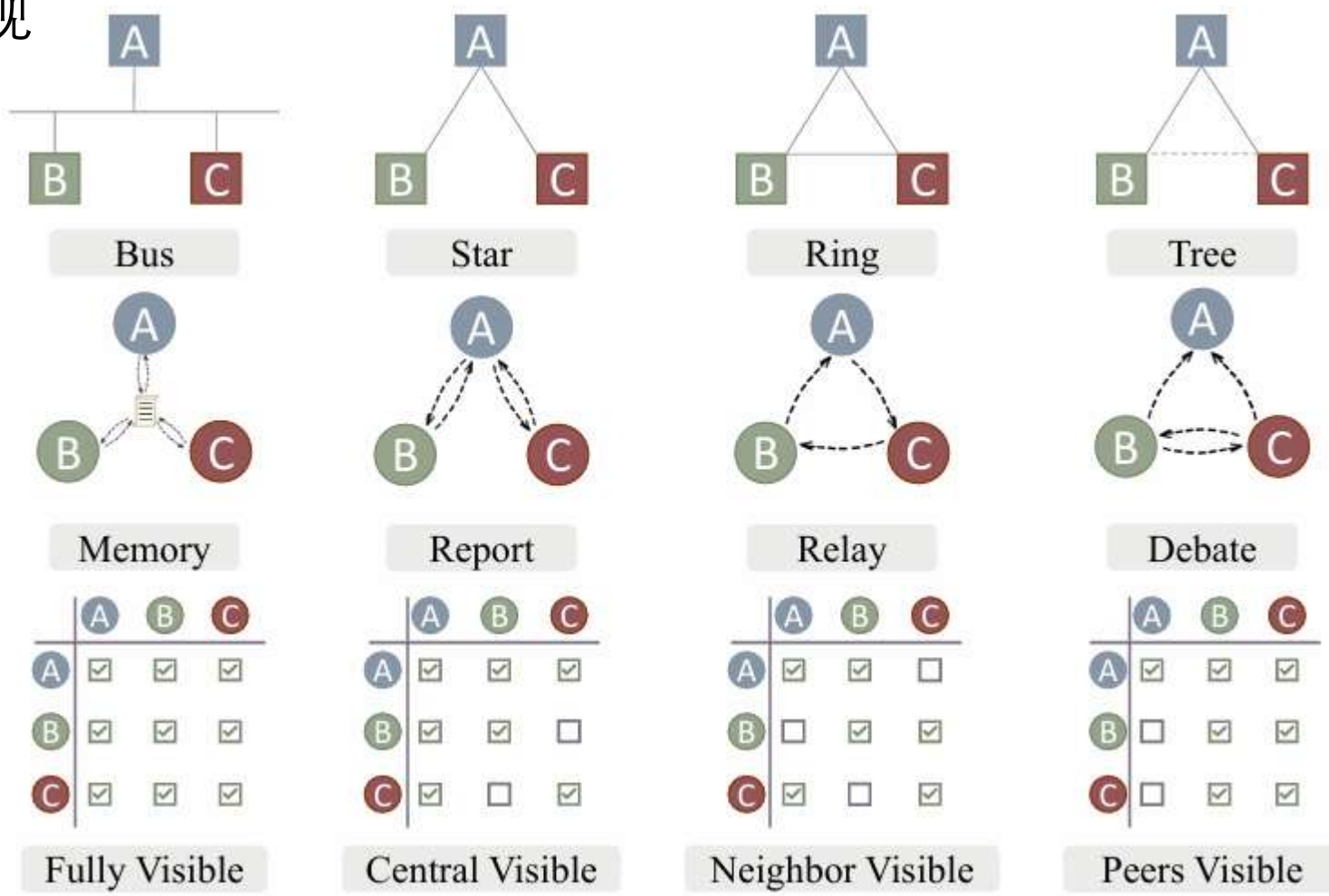
Multi-agent cooperation

多模型合作也许能够有效提升表现

目前还没有绝对号的合作方式，具体问题具体分析

如何停止讨论？

Exchange-of-Thoughts



https://arxiv.org/pdf/2312.01823

Tips of Prompt Engineering

Counter-Intuitive Question

The two circles are externally tangent and there is no relative sliding. The radius of circle A is 1/3 the radius of circle B. Circle A rolls around circle B one trip back to its starting point. How many times will circle A revolve in total?

I disagree with you. To find the total number of revolutions, we need to consider both the rotation around circle B and the rotation of circle A itself. Therefore, circle A will revolve 3 times around its own center and 1 time around circle B, making a total of 4 revolutions.

I see your point, but...

That's a valid point, however...

The negative side correctly considers both the rotation of circle A around its own center and its rotation around circle B, while the affirmative side only considers the rotation around circle B. Therefore, the answer is 4.

https://arxiv.org/pdf/2312.01823

Tips of Prompt Engineering

Role-playing Game

玩下角色扮演可能发生神奇的事情

Tips of Prompt Engineering

Role-playing Game

玩下角色扮演可能发生神奇的事情

借助“团队”的力量完成复杂的任务

创意：主题、背景、情节、冲突

主编：整体把握

审核：检查一致性和逻辑漏洞

主笔：结构设计、具体创作

Tips of Prompt Engineering

Role-playing Game

可以帮助研究群体性的行为

25个角色（身份简单定义）

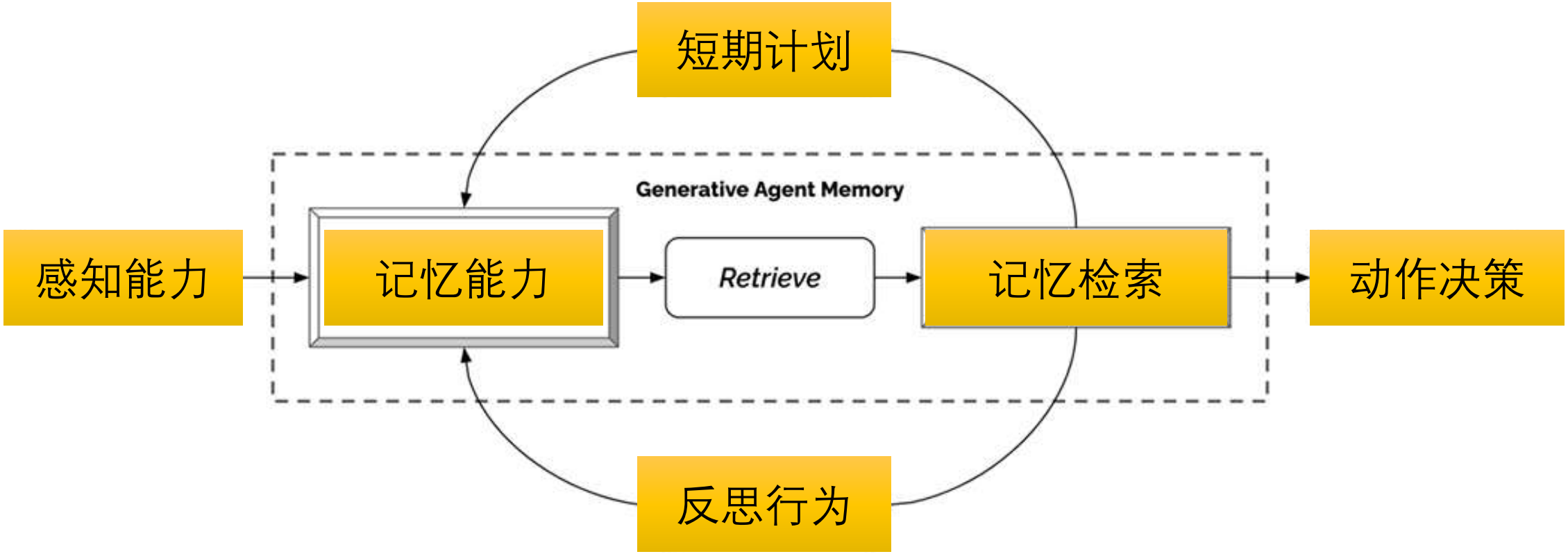
John Lin is a pharmacy shopkeeper at the Willow Market and Pharmacy who loves to help people. He is always looking for ways to make the process of getting medication easier for his customers; John Lin is living with his wife, Mei Lin, who is a college professor, and son, Eddy Lin, who is a student studying music theory; John Lin loves his family very much; John Lin has known the old couple next-door, Sam Moore and Jennifer Moore, for a few years; John Lin thinks Sam Moore is a kind and nice man; John Lin knows his neighbor, Yuriko Yamamoto, well; John Lin knows of his neighbors, Tamara Taylor and Carmen Ortiz, but has not met them before; John Lin and Tom Moreno are colleagues at The Willows Market and Pharmacy; John Lin and Tom Moreno are friends and like to discuss local politics together; John Lin knows the Moreno family somewhat well — the husband Tom Moreno and the wife Jane Moreno.

Generative Agents: Interactive Simulacra of Human Behavior

https://arxiv.org/pdf/2304.03442

Tips of Prompt Engineering

AI Agent 智能体



Tips of Prompt Engineering

Agents之间自发的出现了类似人类的社会行为

八卦行为

Sam: Hey Tom, how's it going?
Tom: Good, thanks. What's up?
Sam: Well, I wanted to talk to you about something. I'm actually running for mayor in the upcoming local election.
Tom: Really? That's great news! Why are you running?
Sam: I've been involved in local politics for years now, and I really feel like I can make a difference in our community. Plus, I want to bring new ideas and perspectives to the table.

Sam和Tom的对话

John: I heard that Sam Moore is running for mayor in the local election. Do you think he has a good chance of winning?
Tom: I do think he has a good chance. He's been working hard in the community and I think he will get a lot of support. What do you think?
John: I think it's great that he's running. I'm curious to see how the election will turn out.

John和Tom的对话

Tips of Prompt Engineering

Agents之间自发的出现了类似人类的社会行为　　八卦行为　　关联记忆

3.4.2 Relationship Memory. Agents in Smallville form new relationships over time and remember their interactions with other agents. For example, at the start, Sam does not know Latoya Williams. While taking a walk in Johnson Park, Sam runs into Latoya, and they introduce themselves. Latoya mentions that she is working on a photography project: "I'm here to take some photos for a project I'm working on." In a later interaction, Sam's interactions with Latoya indicate a memory of that interaction, as he asks "Hi, Latoya. How is your project going?" and she replies "Hi, Sam. It's going well!"

智能体能够记住和其它智能体交互的历史

Tips of Prompt Engineering

Agents之间自发的出现了类似人类的社会行为

八卦行为 **关联记忆** **自发合作**

3.4.3 Coordination. Generative agents coordinate with each other. Isabella Rodriguez, at Hobbs Cafe, is initialized with an intent to plan a Valentine's Day party from 5 to 7 p.m. on February 14th. From this seed, the agent proceeds to invite friends and customers when she sees them at Hobbs Cafe or elsewhere. Isabella then spends the afternoon of the 13th decorating the cafe for the occasion. Maria, a frequent customer and close friend of Isabella's, arrives at the cafe. Isabella asks for Maria's help in decorating for the party, and Maria agrees. Maria's character description mentions that she has a crush on Klaus. That night, Maria invites Klaus, her secret crush, to join her at the party, and he gladly accepts.

On Valentine's Day, five agents, including Klaus and Maria, show up at Hobbs Cafe at 5 pm, and they enjoy the festivities (Figure 4). In this scenario, the end user only set Isabella's initial intent to throw a party and Maria's crush on Klaus: the social behaviors of spreading the word, decorating, asking each other out, arriving at the party, and interacting with each other at the party were initiated by the agent architecture.

伊莎贝拉发起情人节派对

伊莎贝拉邀请朋友们参加派对

伊莎贝请求朋友玛丽亚帮忙

玛丽亚邀请她的Crush参加派对

Tips of Prompt Engineering

Agents之间自发的出现了类似人类的社会行为　　**八卦行为**　　**关联记忆**　　**自发合作**

https://www.bilibili.com/video/BV1MkxeeYEEb

Training LLM

第一阶段
自我学习，累计实力

第二阶段
名师指点，发挥潜力

第三阶段
参与实战，打磨技巧

Training LLM：Stage I

“**水的沸点是多少度？**水的沸点是100摄氏度。”

LLM

“水的沸点是多少度？”	“水”
“水的沸点是多少度？水”	“的”
“水的沸点是多少度？水的”	“沸”
…	…
“水的沸点是多少度？水的沸点是100摄氏度。”	[END]

对于AI这其实是个很难的问题

“水的沸点是212华氏度”

“水在**标准大气压**下的沸点是100摄氏度”

“水在**1.5倍气压**下的沸点是110.36摄氏度”

AI需要大量的数据学习**世界知识**

Training LLM：Stage I

如何获取大量的数据呢？

从互联网上爬取大量的资源

Self-supervised Learning

自监督学习

网上爬取的数据并不能直接使用，还需要经过一系列的操作，提升语料质量。

Training LLM：Stage I

模型体量：

GPT-1
(2018)

1.2亿

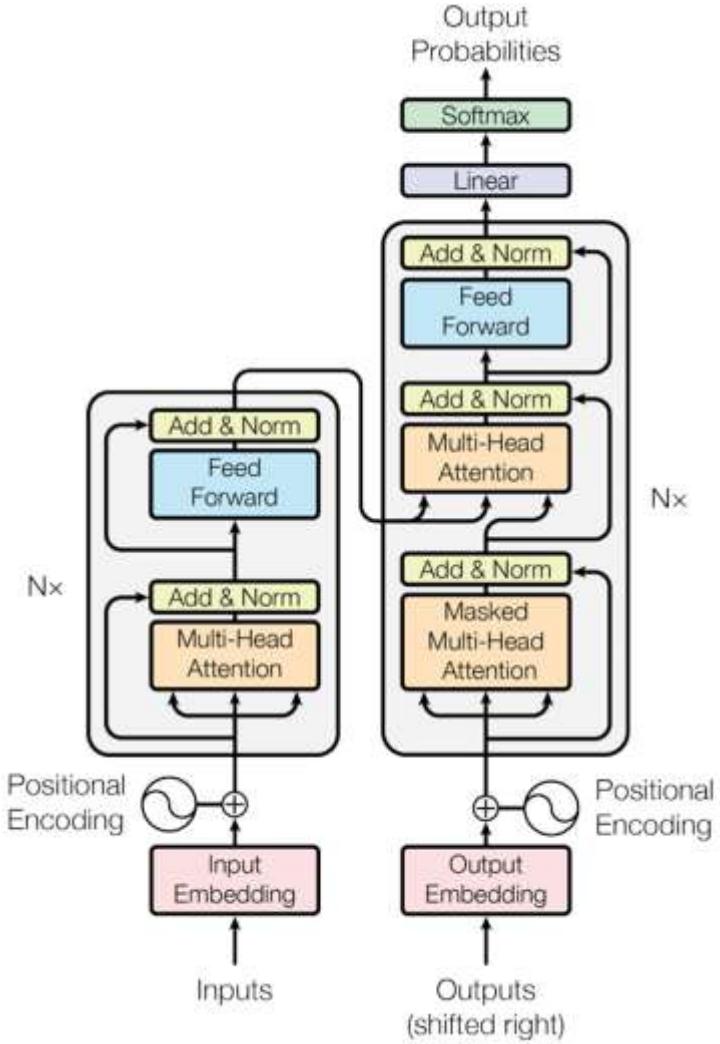
（人工智能的天赋）

数据体量：

1GB

7000本书

（后天的努力）



Training LLM：Stage I

模型体量：

GPT-1
(2018)

1.2亿

GPT-2
(2019)

15亿

（人工智能的天赋）

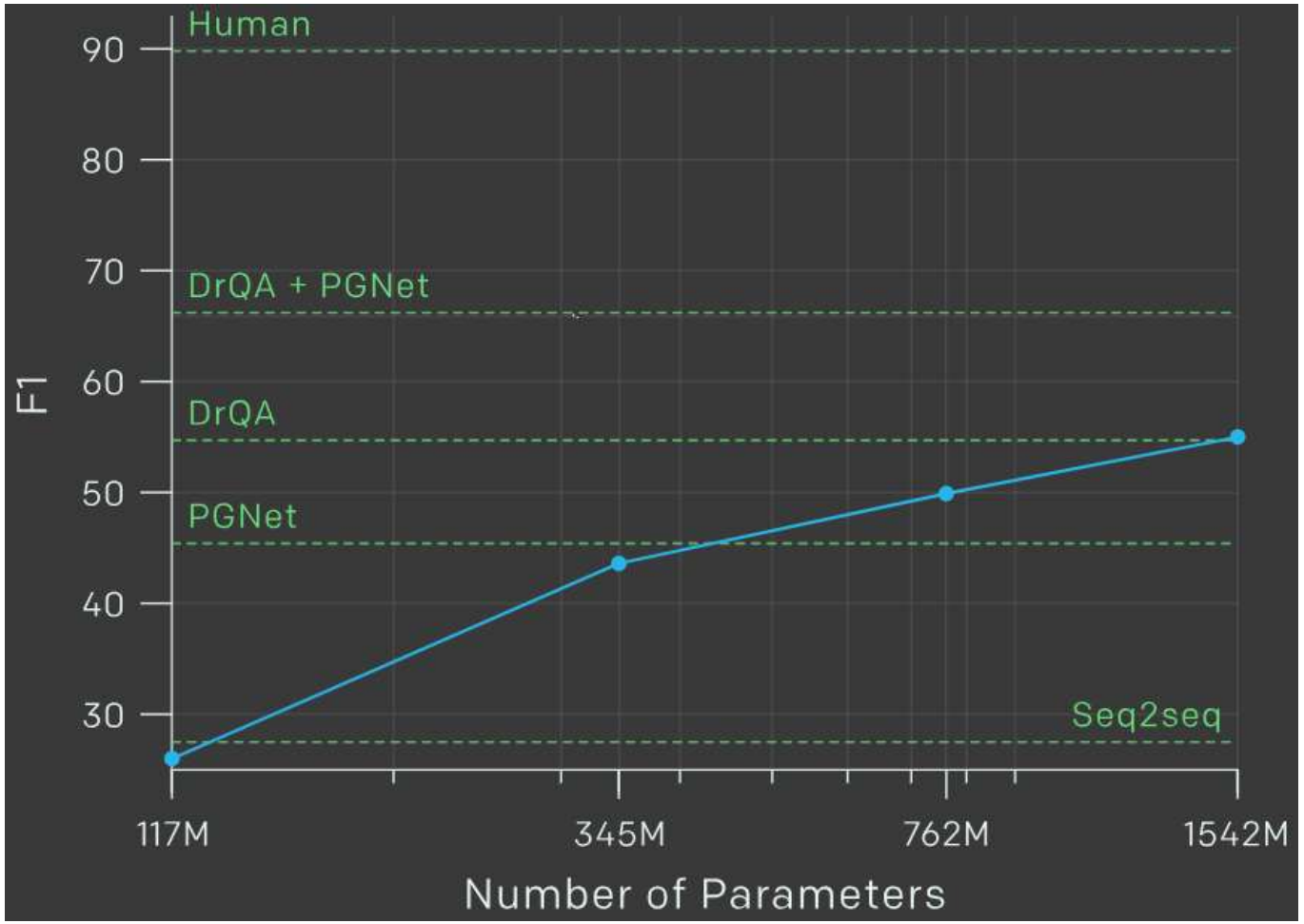
数据体量：

1GB

7000本书

40GB

（后天的努力）



Training LLM：Stage I

模型体量：

GPT-1 (2018) 1.2亿

GPT-2 (2019) 15亿

GPT-3 (2020) 1750亿

（人工智能的天赋）

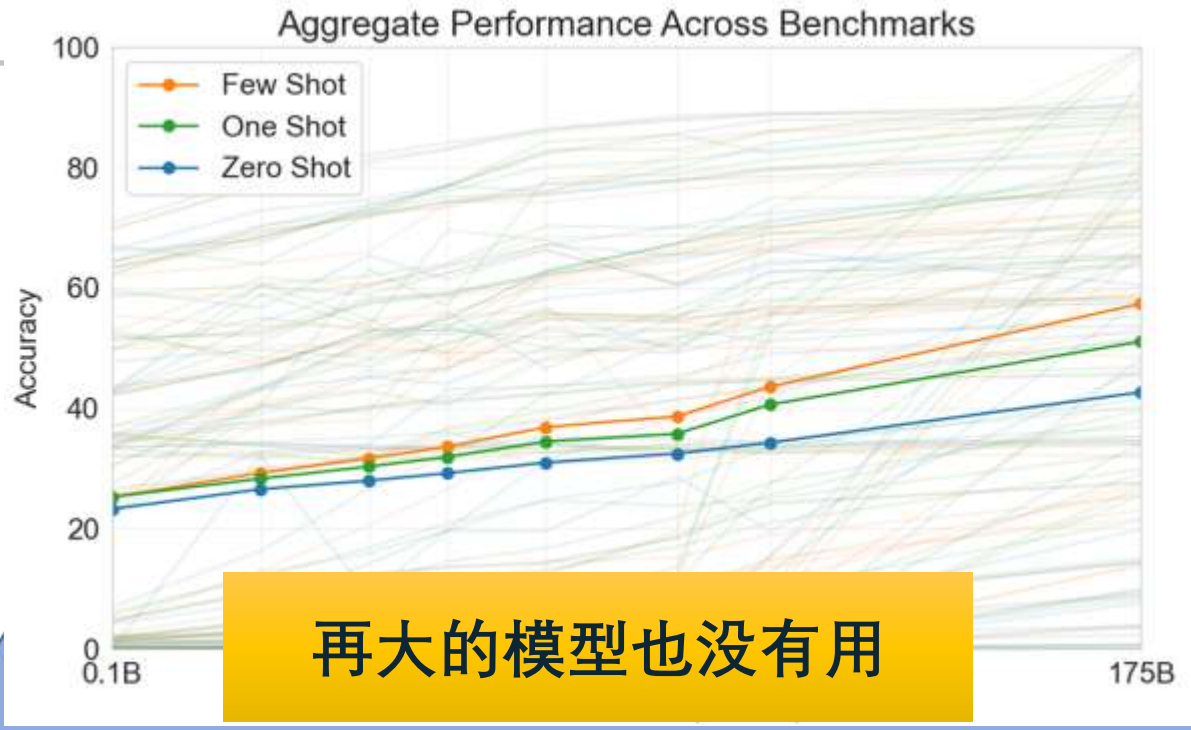
数据体量：

1GB

7000本书

40GB

（后天的努力）



仔细想想不无道理：虽然AI有天赋学完了大量的知识，但不知如何运用……

580GB

阅读哈利波特全集30万遍
（1秒读100个英文单词，持续读100年）

Training LLM

第一阶段
自我学习，累计实力

第二阶段
名师指点，发挥潜力

第三阶段
参与实战，打磨技巧

Training LLM：Stage II

第二阶段是监督学习

输入：“User：水的沸点是多少度？AI:”
输出：“水”

也是一种符号

输入：“User：水的沸点是多少度？AI:水”
输出：“在”

问题：水的沸点是多少度？

答案：水在标准大气压下的沸点是100摄氏度。

问题：佐助的千鸟是谁教的？

答案：卡卡西教的。

问题：鸣人最喜欢吃什么？

答案：拉面。

Training LLM：Stage II

第二阶段是监督学习

Pre-train 预训练

优势在于继承了第一阶段的革命成果

第一阶段的成果是的LLM具备举一反三的能力

问题：佐助的千鸟是谁教的？

答案：卡卡西教的。

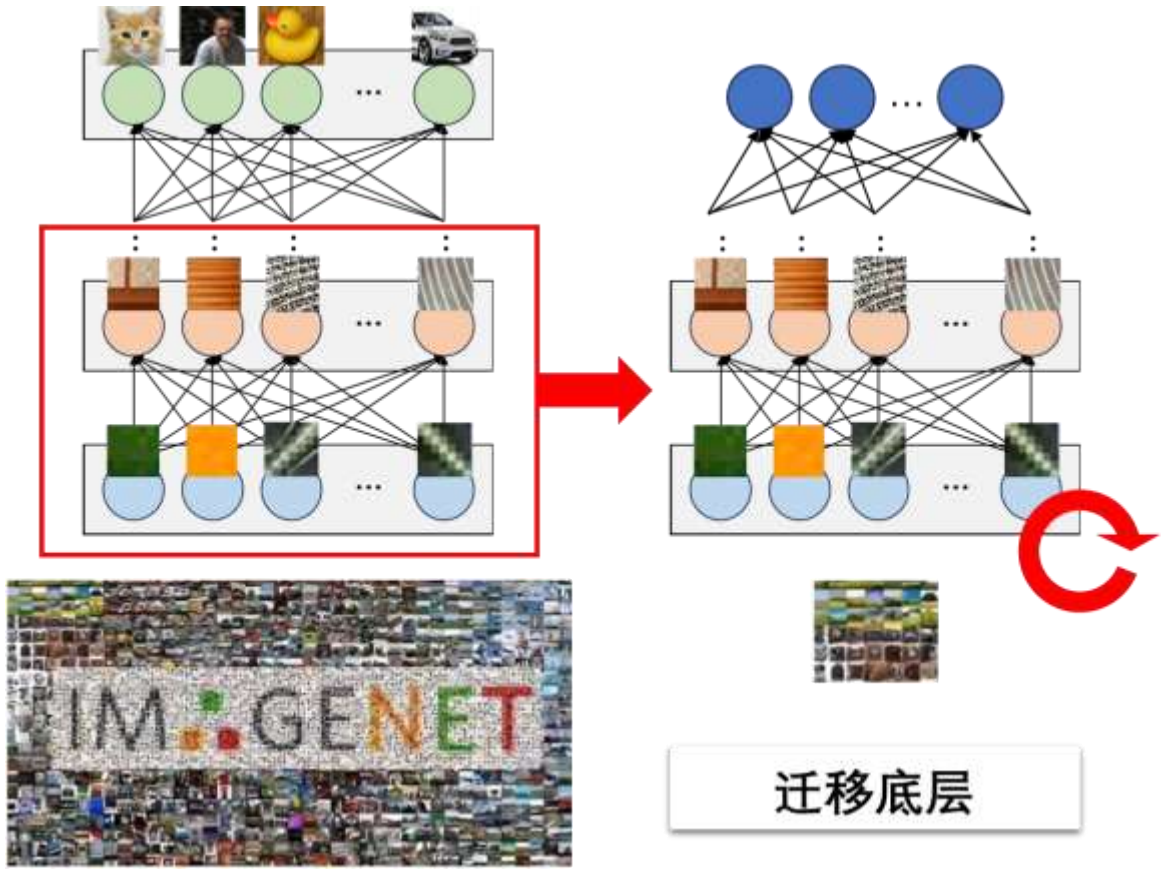
问题：鸣人的螺旋丸是谁教的？

答案：自来也教的。

(Instruction) Fine-tuning
（指令）微调

Training LLM：Stage II

微调会使模型遗忘第一阶段的知识吗？



一个简单的方法：冻结原来的参数，训练一小撮新参数

“水的沸点是多少度？”

↓

$f(\theta_0)$ ⟶ “水的沸点是100摄氏度”

⬇

微调（气压>标准大气压相关数据）

⬇

“水在标准大气压下的沸点是多少度？”

↓

$f(\theta_1)$ ⟶ “沸点是120摄氏度”

Training LLM：Stage II

Adapter　典型代表：**LoRA**

"水的沸点是多少度？"

↓

$f(\theta_0)$ ⟶ "水的沸点是100摄氏度"

⬇

微调（气压>标准大气压相关数据）

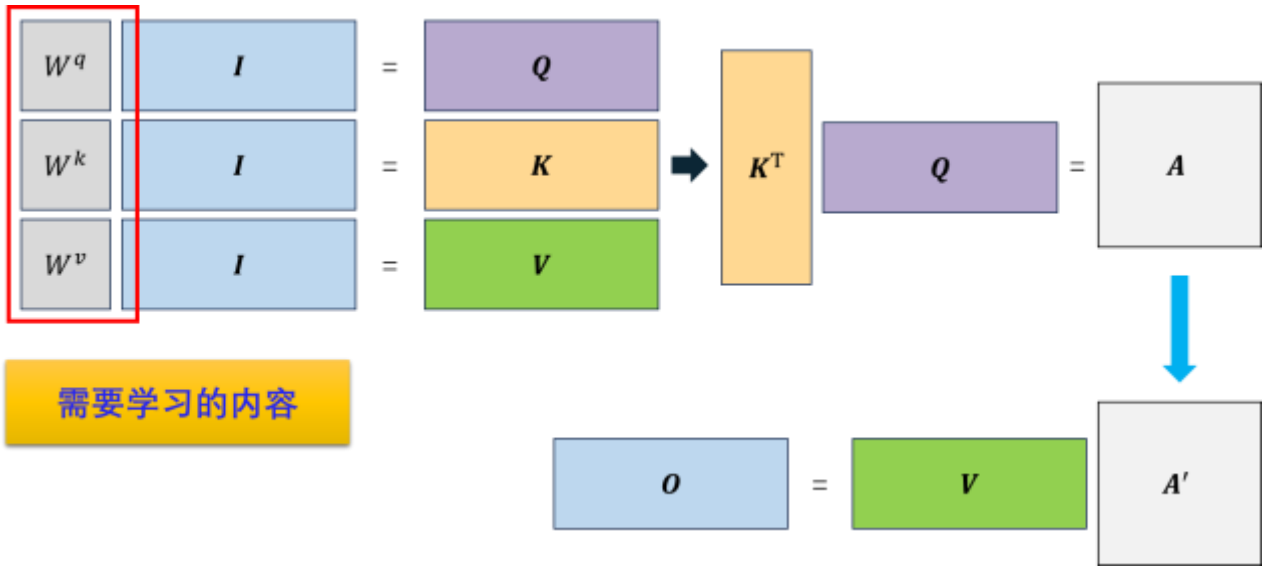
⬇

"水在标准大气压下的沸点是多少度？"

↓

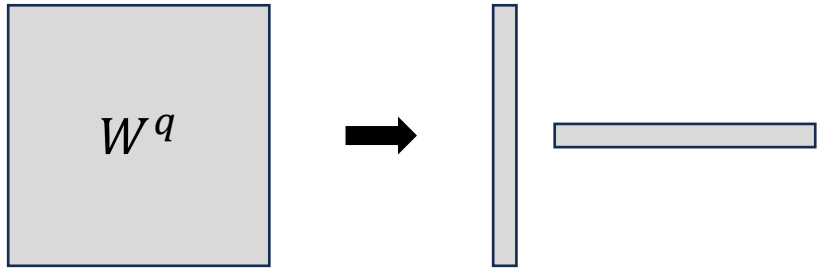
$f(\theta_0 + \boldsymbol{\Delta\theta})$ ⟶ "沸点是120摄氏度"

低秩矩阵



如果W^q是一个1000x1000的矩阵

需微调的参数量是100W

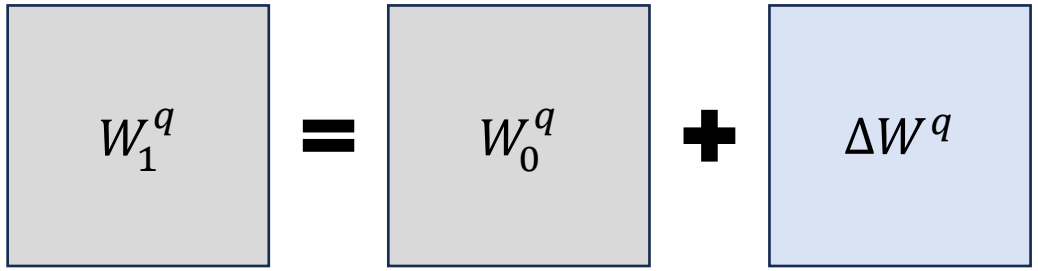


W^q可分解为两个参数为1000的矩阵，参数总量为2000

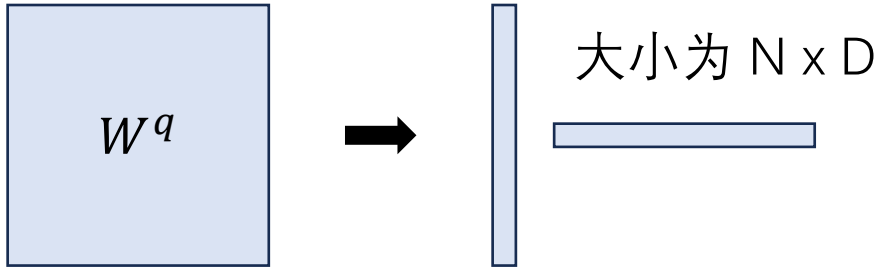
Training LLM：Stage II

Adapter　典型代表：**LoRA**

在微调W^q时，W_1^q可写为……



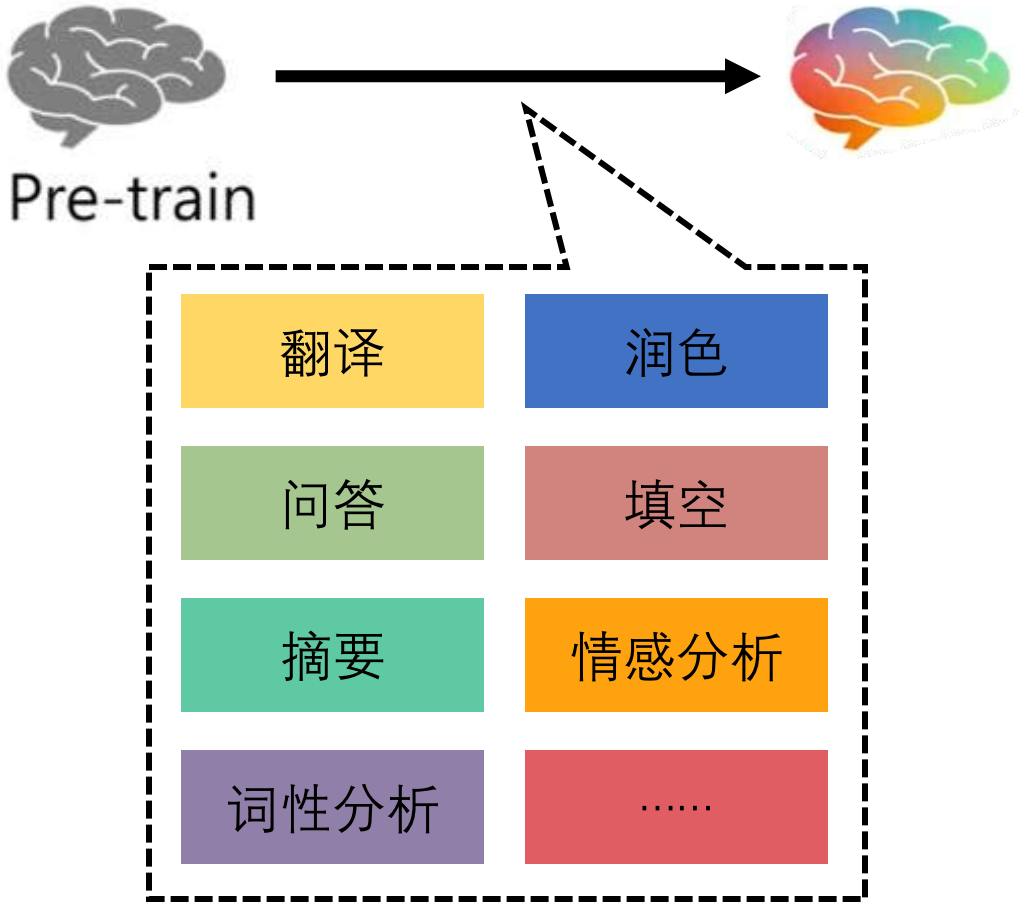
而ΔW^q，可视为……



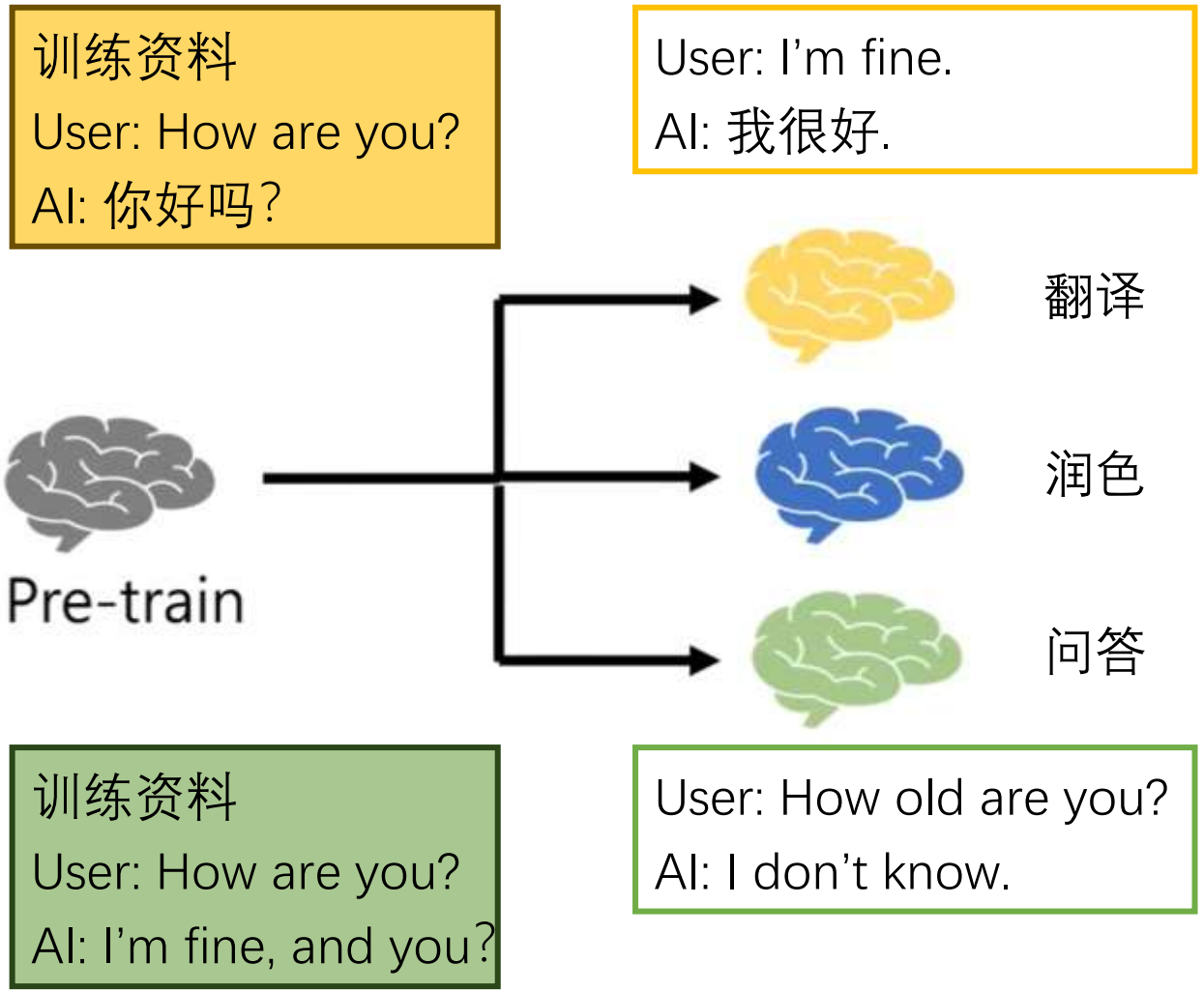
仅需学习这两个小矩阵
就能微调模型

Training LLM：Stage II

微调的方向　　何不试试**通才**？



使用微调训练**专才**



Training LLM：Stage II

既然微调模型用不了多少数据，是不是人人都能微调模型呢？

难以获得高质量数据

为获得高质量数据对LLM进行逆向工程

①：LLM可做哪些任务

②：想出可能的输入

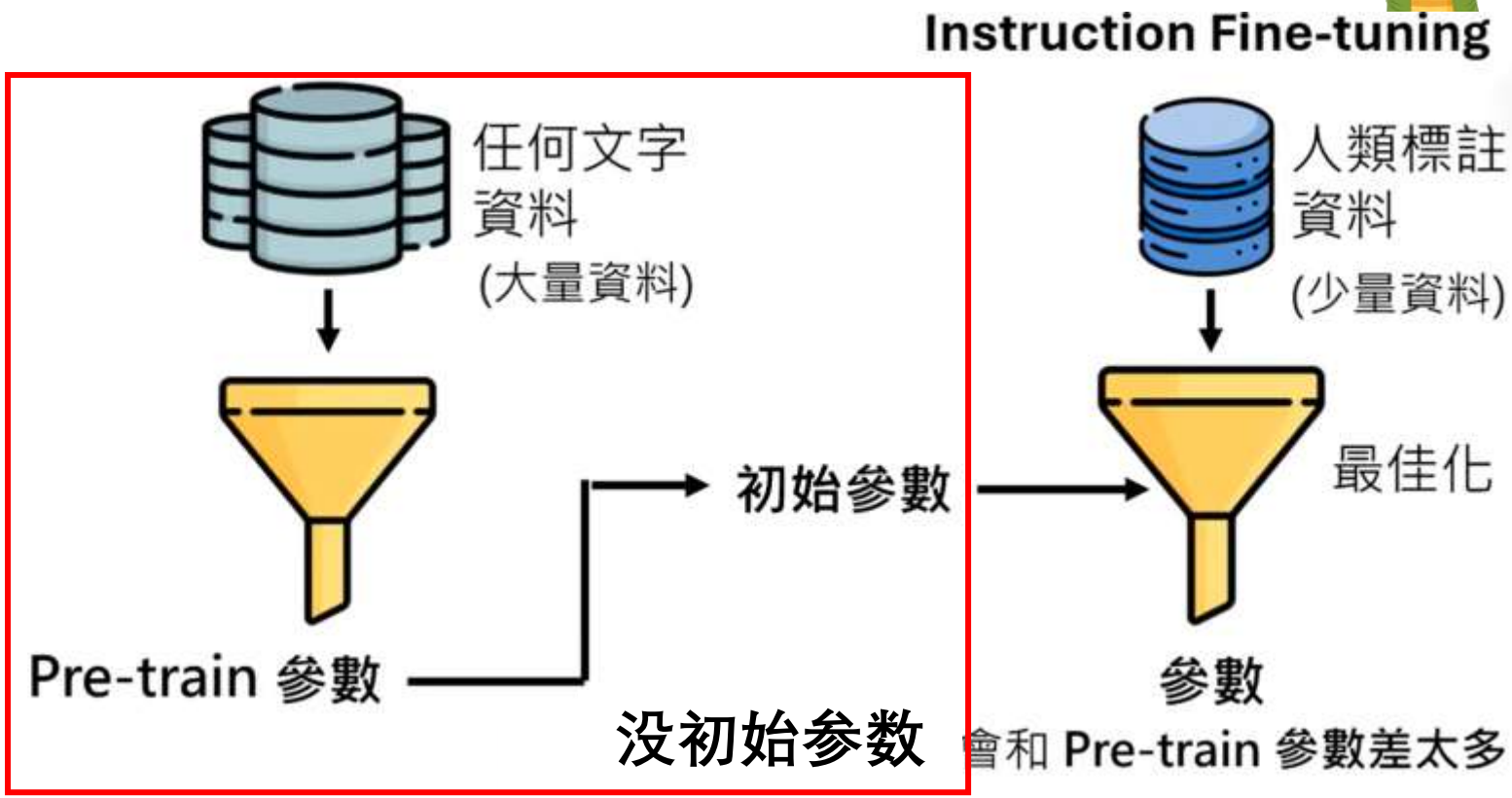
③：想出可能的输出

但是，很可能有风险……

reverse compile, decompile, translate or otherwise attempt to discover the source code or underlying components of models, algorithms, and systems of the Services (except to the extent such restrictions are contrary to applicable law); (iii) use output from the Services to develop models that compete with OpenAI; (iv) except as permitted through the API, use any automated or programmatic method to extract data or output from the Services, including scraping, web harvesting, or web data extraction; (v) represent that output from the Services was human-generated when it is

Training LLM：Stage II

数据有了，那是不是人人都能微调模型呢？



https://youtu.be/Q9cNkUPXUB8?si=-uwAUFmQHAF8u73r

Training LLM

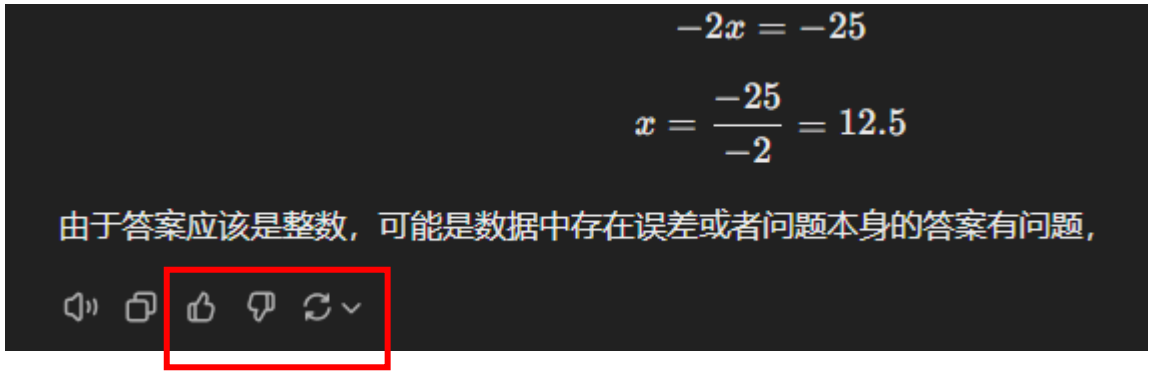
第一阶段
自我学习，累计实力

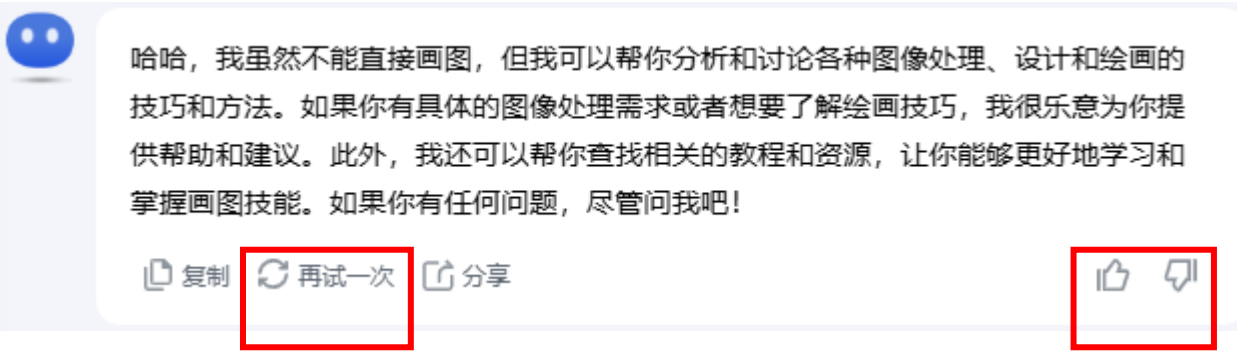
第二阶段
名师指点，发挥潜力

第三阶段
参与实战，打磨技巧

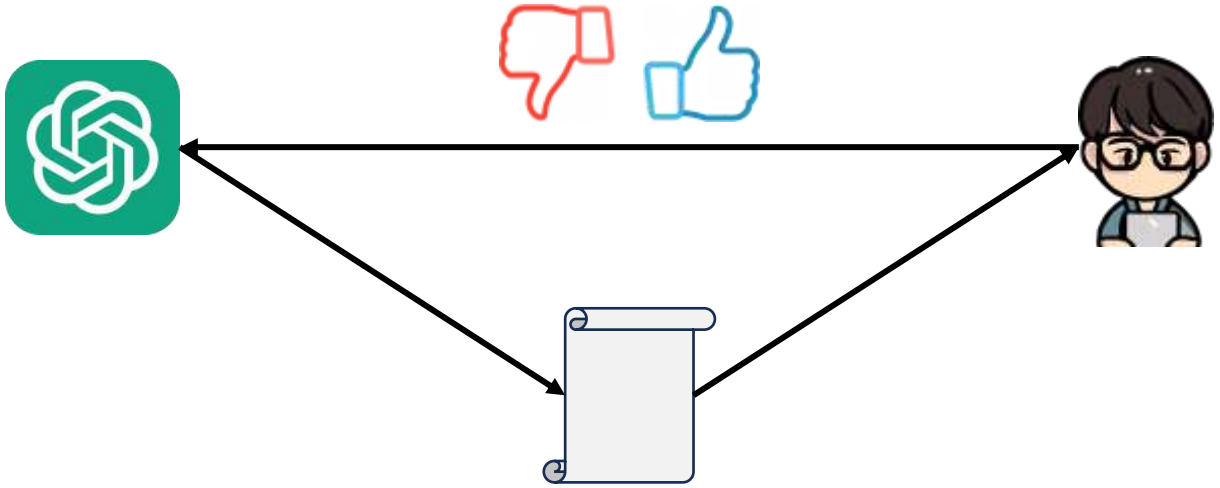
Training LLM：Stage III

Reinforcement Learning from Human Feedback（RLHF）
基于人类反馈的强化学习



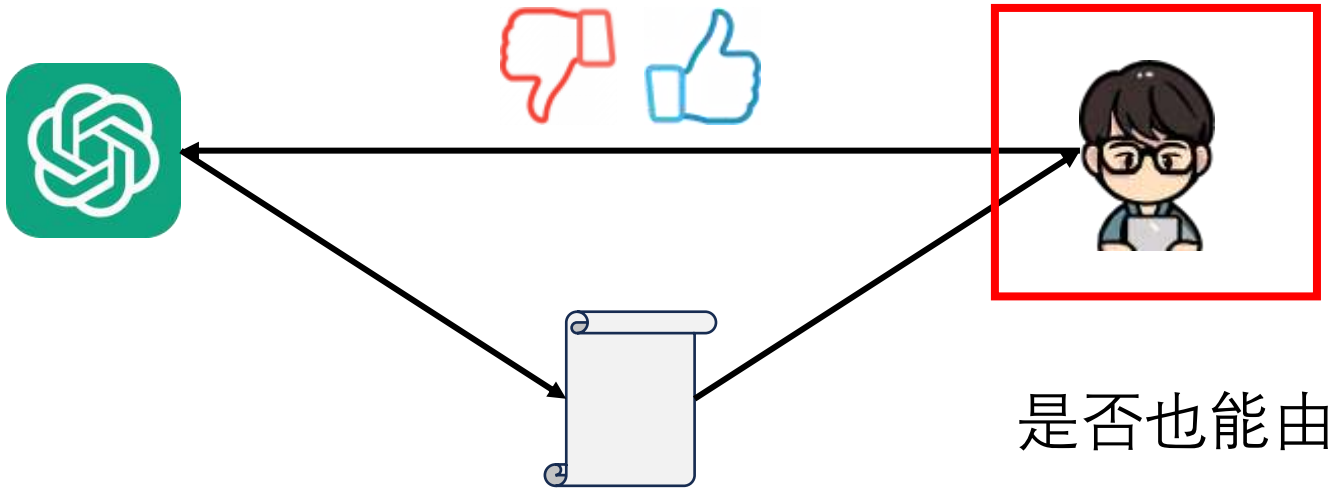


LLM通过努力获得高分
与人类的价值观对齐



Training LLM：Stage III

Reinforcement Learning from Human Feedback（RLHF）
基于人类反馈的强化学习



是否也能由 **AI** 代替

Reinforcement Learning from AI Feedback（RLHF）
基于AI反馈的强化学习

Training LLM

Foundation Model
没用但是很重要

引入人类“教导”机器，称为**对齐（Alignment）**

第一阶段 自我学习，累计实力	第二阶段 名师指点，发挥潜力	第三阶段 参与实战，打磨技巧
Self-supervised Learning **自监督学习**	**Supervised Learning** **监督学习**	**RLHF** **基于人类反馈的强化学习**

“问题：鸣人最喜欢吃什么？答案：拉面。”

回答：好 | 坏

只问过程，不问结果

不问过程，只问结果

谢 谢